



- 3.9 Appropriate notification to regulatory authorities should be made where significant data integrity incidents have been identified.
- 3.10 The guidance refers to the acronym ALCOA rather than 'ALCOA+'. ALCOA being Attributable, Legible, Contemporaneous, Original, and Accurate and the '+' referring to Complete, Consistent, Enduring, and Available. ALCOA was historically regarded as defining the attributes of data quality that are suitable for regulatory purposes. The '+' has been subsequently added to emphasise the requirements. There is no difference in expectations regardless of which acronym is used since data governance measures should ensure that data is complete, consistent, enduring and available throughout the data lifecycle.

4. Establishing data criticality and inherent integrity risk

- 4.1 Data has varying importance to quality, safety and efficacy decisions. Data criticality may be determined by considering how the data is used to influence the decisions made.
- 4.2 The risks to data are determined by the potential to be deleted, amended or excluded without authorisation and the opportunity for detection of those activities and events. The risks to data may be increased by complex, inconsistent processes with open-ended and subjective outcomes, compared to simple tasks that are undertaken consistently, are well defined and have a clear objective.
- 4.3 Data may be generated by:
- (i) Recording on paper, a paper-based record of a manual observation or of an activity or
 - (ii) electronically, using equipment that range from simple machines through to complex highly configurable computerised systems or
 - (iii) by using a hybrid system where both paper-based and electronic records constitute the original record or
 - (iv) by other means such as photography, imagery, chromatography plates, etc.

Paper

Data generated manually on paper may require independent verification if deemed necessary from the data integrity risk assessment or by another requirement. Consideration should be given to risk-reducing supervisory measures.

Electronic

The inherent risks to data integrity relating to equipment and computerised systems may differ depending upon the degree to which the system generating or using the data can be configured, and the potential for manipulation of data during transfer between computerised systems during the data lifecycle.

The use of available technology, suitably configured to reduce data integrity risk, should be considered.



Simple electronic systems with no configurable software and no electronic data retention (e.g. pH meters, balances and thermometers) may only require calibration, whereas complex systems require 'validation for intended purpose'.

Validation effort increases with complexity and risk (determined by software functionality, configuration, the opportunity for user intervention and data lifecycle considerations). It is important not to overlook systems of apparent lower complexity. Within these systems, it may be possible to manipulate data or repeat testing to achieve the desired outcome with limited opportunity for detection (e.g. stand-alone systems with a user-configurable output such as ECG machines, FTIR, UV spectrophotometers).

Hybrid

Where hybrid systems are used, it should be clearly documented what constitutes the whole data set and all records that are defined by the data set should be reviewed and retained. Hybrid systems should be designed to ensure they meet the desired objective.

Other

Where the data generated is captured by a photograph or imagery (or other media), the requirements for storage of that format throughout its lifecycle should follow the same considerations as for the other formats, considering any additional controls required for that format. Where the original format cannot be retained due to degradation issues, alternative mechanisms for recording (e.g. photography or digitisation) and subsequent storage may be considered and the selection rationale documented (e.g. thin layer chromatography).

- 4.4 Reduced effort and/or frequency of control measures may be justified for data that has a lesser impact to product, patient or the environment if those data are obtained from a process that does not provide the opportunity for amendment without high-level system access or specialist software/knowledge.
- 4.5 The data integrity risk assessment (or equivalent) should consider factors required to follow a process or perform a function. It is expected to consider not only a computerised system but also the supporting people, guidance, training and quality systems. Therefore, automation or the use of a 'validated system' (e.g. e-CRF; analytical equipment) may lower but not eliminate data integrity risk. Where there is human intervention, particularly influencing how or what data is recorded, reported or retained, an increased risk may exist from poor organisational controls or data verification due to an overreliance on the system's validated state.
- 4.6 Where the data integrity risk assessment has highlighted areas for remediation, prioritisation of actions (including acceptance of an appropriate level of residual risk) should be documented, communicated to management, and subject to review. In situations where long-term remediation actions are identified, risk-reducing short-term measures should be implemented to provide acceptable data governance in the interim.



5. Designing systems and processes to assure data integrity; creating the 'right environment'.

5.1 Systems and processes should be designed in a way that facilitates compliance with the principles of data integrity. Enablers of the desired behaviour include but are not limited to:

- At the point of use, having access to appropriately controlled/synchronised clocks for recording timed events to ensure reconstruction and traceability, knowing and specifying the time zone where this data is used across multiple sites.
- Accessibility of records at locations where activities take place so that informal data recording and later transcription to official records does not occur.
- Access to blank paper proformas for raw/source data recording should be appropriately controlled. Reconciliation, or the use of controlled books with numbered pages, may be necessary to prevent recreation of a record. There may be exceptions such as medical records (GCP) where this is not practical.
- User access rights that prevent (or audit trail, if prevention is not possible) unauthorised data amendments. Use of external devices or system interfacing methods that eliminate manual data entries and human interaction with the computerised system, such as barcode scanners, ID card readers, or printers.
- The provision of a work environment (such as adequate space, sufficient time for tasks, and properly functioning equipment) that permit performance of tasks and recording of data as required.
- Access to original records for staff performing data review activities.
- Reconciliation of controlled print-outs.
- Sufficient training in data integrity principles provided to all appropriate staff (including senior management).
- Inclusion of subject matter experts in the risk assessment process.
- Management oversight of quality metrics relevant to data governance.

5.2 The use of scribes to record activity on behalf of another operator can be considered where justified, for example:

- The act of contemporaneous recording compromises the product or activity e.g. documenting line interventions by sterile operators.
- Necropsy (GLP)
- To accommodate cultural or literacy/language limitations, for instance where an activity is performed by an operator but witnessed and recorded by a second person.

Consideration should be given to ease of access, usability and location whilst ensuring appropriate control of the activity guided by the criticality of the data.

In these situations, the recording by the second person should be contemporaneous with the task being performed, and the records should identify both the person performing the task and the person completing the record. The person performing the task should countersign the record wherever possible, although it is accepted that this countersigning step will be retrospective. The process for supervisory (scribe) documentation completion should be described in an approved procedure that specifies the activities to which the process applies.



6. Definition of terms and interpretation of requirements

In the following section, definitions where applicable, are given in italic text directly below the term.

6.1. Data

Facts, figures and statistics collected together for reference or analysis. All original records and true copies of original records, including source data and metadata and all subsequent transformations and reports of these data, that are generated or recorded at the time of the GXP activity and allow full and complete reconstruction and evaluation of the GXP activity.

Data should be:

- A - attributable to the person generating the data
- L – legible and permanent
- C – contemporaneous
- O – original record (or certified true copy)
- A - accurate

Data governance measures should also ensure that data is complete, consistent, enduring and available throughout the lifecycle, where;

Complete – the data must be whole; a complete set

Consistent - the data must be self-consistent

Enduring – durable; lasting throughout the data lifecycle

Available – readily available for review or inspection purposes

6.2. Raw data (synonymous with ‘source data’ which is defined in ICH GCP)

Raw data is defined as the original record (data) which can be described as the first-capture of information, whether recorded on paper or electronically. Information that is originally captured in a dynamic state should remain available in that state.

Raw data must permit full reconstruction of the activities. Where this has been captured in a dynamic state and generated electronically, paper copies cannot be considered as ‘raw data’.

In the case of basic electronic equipment that does not store electronic data, or provides only a printed data output (e.g. balances or pH meters), then the printout constitutes the raw data. Where the basic electronic equipment does store electronic data permanently and only holds a certain volume before overwriting; this data should be periodically reviewed and where necessary reconciled against paper records and extracted as electronic data where this is supported by the equipment itself.

In all definitions, the term 'data' includes raw data.



6.3. Metadata

Metadata are data that describe the attributes of other data and provide context and meaning. Typically, these are data that describe the structure, data elements, inter-relationships and other characteristics of data e.g. audit trails. Metadata also permit data to be attributable to an individual (or if automatically generated, to the original data source).

Metadata form an integral part of the original record. Without the context provided by metadata the data has no meaning.

Example (i) 3.5

metadata, giving context and meaning, (*italic text*) are:

sodium chloride batch 1234, 3.5mg. J Smith 01/Jul/14

Example (ii) 3.5

metadata, giving context and meaning, (*italic text*) are:

*Trial subject A123, sample ref X789 taken 30/06/14 at 1456hrs.
3.5mg. Analyst: J Smith 01/Jul/14*

6.4. Data Integrity

Data integrity is the degree to which data are complete, consistent, accurate, trustworthy, reliable and that these characteristics of the data are maintained throughout the data life cycle. The data should be collected and maintained in a secure manner, so that they are attributable, legible, contemporaneously recorded, original (or a true copy) and accurate. Assuring data integrity requires appropriate quality and risk management systems, including adherence to sound scientific principles and good documentation practices.

6.5. Data Governance

The arrangements to ensure that data, irrespective of the format in which they are generated, are recorded, processed, retained and used to ensure the record throughout the data lifecycle.

Data governance should address data ownership and accountability throughout the lifecycle, and consider the design, operation and monitoring of processes/systems to comply with the principles of data integrity including control over intentional and unintentional changes to data.

Data Governance systems should include staff training in the importance of data integrity principles and the creation of a working environment that enables visibility, and actively encourages reporting of errors, omissions and undesirable results.

Senior management should be accountable for the implementation of systems and procedures to minimise the potential risk to data integrity, and for identifying the residual risk, using risk management techniques such as the principles of ICH Q9.



Contract Givers should ensure that data ownership, governance and accessibility are included in any contract/technical agreement with a third party. The Contract Giver should also perform a data governance review as part of their vendor assurance programme.

Data governance systems should also ensure that data are readily available and directly accessible on request from national competent authorities. Electronic data should be available in human-readable form.

6.6. Data Lifecycle

All phases in the life of the data from generation and recording through processing (including analysis, transformation or migration), use, data retention, archive/retrieval and destruction.

Data governance, as described in the previous section, must be applied across the whole data lifecycle to provide assurance of data integrity. Data can be retained either in the original system, subject to suitable controls, or in an appropriate archive.

6.7. Recording and collection of data

No definition required.

Organisations should have an appropriate level of process understanding and technical knowledge of systems used for data collection and recording, including their capabilities, limitations and vulnerabilities.

The selected method should ensure that data of appropriate accuracy, completeness, content and meaning are collected and retained for their intended use. Where the capability of the electronic system permits dynamic storage, it is not appropriate for static (printed / manual) data to be retained in preference to dynamic (electronic) data.

As data are required to allow the full reconstruction of activities the amount and the resolution (degree of detail) of data to be collected should be justified.

When used, blank forms (including, but not limited to, worksheets, laboratory notebooks, and master production and control records) should be controlled. For example, numbered sets of blank forms may be issued and reconciled upon completion. Similarly, bound paginated notebooks, stamped or formally issued by a document control group allow detection of unofficial notebooks and any gaps in notebook pages.

6.8. Data transfer / migration

Data transfer is the process of transferring data between different data storage types, formats, or computerised systems.

Data migration is the process of moving stored data from one durable storage location to another. This may include changing the format of data, but not the content or meaning.

Data transfer is the process of transferring data and metadata between storage media types or computerised systems. Data migration where required may, if necessary, change the format of data to make it usable or visible on an alternative computerised system.



Data transfer/migration procedures should include a rationale, and be robustly designed and validated to ensure that data integrity is maintained during the data lifecycle. Careful consideration should be given to understanding the data format and the potential for alteration at each stage of data generation, transfer and subsequent storage. The challenges of migrating data are often underestimated, particularly regarding maintaining the full meaning of the migrated records.

Data transfer should be validated. The data should not be altered during or after it is transferred to the worksheet or other application. There should be an audit trail for this process. Appropriate Quality procedures should be followed if the data transfer during the operation has not occurred correctly. Any changes in the middle layer software should be managed through appropriate Quality Management Systems.

Electronic worksheets used in automation like paper documentation should be version controlled and any changes in the worksheet should be documented/verified appropriately.

6.9. Data Processing

A sequence of operations performed on data to extract, present or obtain information in a defined format. Examples might include: statistical analysis of individual patient data to present trends or conversion of a raw electronic signal to a chromatogram and subsequently a calculated numerical result

There should be adequate traceability of any user-defined parameters used within data processing activities to the raw data, including attribution to who performed the activity.

Audit trails and retained records should allow reconstruction of all data processing activities regardless of whether the output of that processing is subsequently reported or otherwise used for regulatory or business purposes. If data processing has been repeated with progressive modification of processing parameters this should be visible to ensure that the processing parameters are not being manipulated to achieve a more desirable result.

6.10. Excluding Data (not applicable to GPvP):

Note: this is not applicable to GPvP; for GPvP refer to the pharmacovigilance legislation (including the GVP modules) which provide the necessary requirements and statutory guidance.

Data may only be excluded where it can be demonstrated through valid scientific justification that the data are not representative of the quantity measured, sampled or acquired.

In all cases, this justification should be documented and considered during data review and reporting. All data (even if excluded) should be retained with the original data set, and be available for review in a format that allows the validity of the decision to exclude the data to be confirmed.

6.11. Original record and true copy

6. 11.1. Original record



The first or source capture of data or information e.g. original paper record of manual observation or electronic raw data file from a computerised system, and all subsequent data required to fully reconstruct the conduct of the GXP activity. Original records can be Static or Dynamic.

A static record format, such as a paper or electronic record, is one that is fixed and allows little or no interaction between the user and the record content. For example, once printed or converted to static electronic format chromatography records lose the capability of being reprocessed or enabling more detailed viewing of baselines.

Records in dynamic format, such as electronic records, allow an interactive relationship between the user and the record content. For example, electronic records in database formats allow the user to track, trend and query data; chromatography records maintained as electronic records allow the user or reviewer (with appropriate access permissions) to reprocess the data and expand the baseline to view the integration more clearly.

Where it is not practical or feasibly possible to retain the original copy of source data, (e.g. MRI scans, where the source machine is not under the study sponsor's control and the operator can only provide summary statistics) the risks and mitigation should be documented.

Where the data obtained requires manual observation to record (for example results of a manual titration, visual interpretation of environmental monitoring plates) the process should be risk assessed and depending on the criticality, justify if a second contemporaneous verification check is required or investigate if the result could be captured by an alternate means.

6.11.2. True copy

A copy (irrespective of the type of media used) of the original record that has been verified (i.e. by a dated signature or by generation through a validated process) to have the same information, including data that describe the context, content, and structure, as the original.

A true copy may be stored in a different electronic file format to the original record if required, but must retain the metadata and audit trail required to ensure that the full meaning of the data are kept and its history may be reconstructed.

Original records and true copies must preserve the integrity of the record. True copies of original records may be retained in place of the original record (e.g. scan of a paper record), if a documented system is in place to verify and record the integrity of the copy. Organisations should consider any risk associated with the destruction of original records.

It should be possible to create a true copy of electronic data, including relevant metadata, for the purposes of review, backup and archival. Accurate and complete copies for certification of the copy should include the meaning of the data (e.g. date formats, context, layout, electronic signatures and authorisations) and the full GXP audit trail. Consideration should be given to the dynamic functionality of a 'true copy' throughout the retention period (see 'archive').

Data must be retained in a dynamic form where this is critical to its integrity or later verification. If the computerised system cannot be maintained e.g., if it is no longer supported, then records should be archived according to a documented archiving strategy prior to



decommissioning the computerised system. It is conceivable for some data generated by electronic means to be retained in an acceptable paper or electronic format, where it can be justified that a static record maintains the integrity of the original data. However, the data retention process must be shown to include verified copies of all raw data, metadata, relevant audit trail and result files, any variable software/system configuration settings specific to each record, and all data processing runs (including methods and audit trails) necessary for reconstruction of a given raw data set. It would also require a documented means to verify that the printed records were an accurate representation. To enable a GXP compliant record this approach is likely to be demanding in its administration.

Where manual transcriptions occur, these should be verified by a second person or validated system.

6.12. Computerised system transactions:

A computerised system transaction is a single operation or sequence of operations performed as a single logical 'unit of work'. The operation(s) that makes a transaction may not be saved as a permanent record on durable storage until the user commits the transaction through a deliberate act (e.g. pressing a save button), or until the system forces the saving of data.

The metadata (e.g. username, date, and time) are not captured in the system audit trail until the user saves the transaction to durable storage. In computerised systems, an electronic signature may be required for the record to be saved and become permanent.

A critical step is a parameter that must be within an appropriate limit, range, or distribution to ensure the safety of the subject or quality of the product or data. Computer systems should be designed to ensure that the execution of critical steps is recorded contemporaneously. Where transactional systems are used, the combination of multiple unit operations into a combined single transaction should be avoided, and the time intervals before saving of data should be minimised. Systems should be designed to require saving data to permanent memory before prompting users to make changes.

The organisation should define during the development of the system (e.g. via the user requirements specification) what critical steps are appropriate based on the functionality of the system and the level of risk associated. Critical steps should be documented with process controls that consider system design (prevention), together with monitoring and review processes. Oversight of activities should alert to failures that are not addressed by the process design.

6.13. Audit Trail

The audit trail is a form of metadata containing information associated with actions that relate to the creation, modification or deletion of GXP records. An audit trail provides for secure recording of life-cycle details such as creation, additions, deletions or alterations of information in a record, either paper or electronic, without obscuring or overwriting the original record. An audit trail facilitates the reconstruction of the history of such events relating to the record regardless of its medium, including the "who, what, when and why" of the action.

Where computerised systems are used to capture, process, report, store or archive raw data electronically, system design should always provide for the retention of audit trails to show all



changes to, or deletion of data while retaining previous and original data. It should be possible to associate all data and changes to data with the persons making those changes, and changes should be dated and time stamped (time and time zone where applicable). The reason for any change, should also be recorded. The items included in the audit trail should be those of relevance to permit reconstruction of the process or activity.

Audit trails (identified by risk assessment as required) should be switched on. Users should not be able to amend or switch off the audit trail. Where a system administrator amends, or switches off the audit trail a record of that action should be retained.

The relevance of data retained in audit trails should be considered by the organisation to permit robust data review/verification. It is not necessary for audit trail review to include every system activity (e.g. user log on/off, keystrokes etc.).

Where relevant audit trail functionality does not exist (e.g. within legacy systems) an alternative control may be achieved for example defining the process in an SOP, and use of log books. Alternative controls should be proven to be effective.

Where add-on software or a compliant system does not currently exist, continued use of the legacy system may be justified by documented evidence that a compliant solution is being sought and that mitigation measures temporarily support the continued use.¹

Routine data review should include a documented audit trail review where this is determined by a risk assessment. When designing a system for review of audit trails, this may be limited to those with GXP relevance. Audit trails may be reviewed as a list of relevant data, or by an 'exception reporting' process. An exception report is a validated search tool that identifies and documents predetermined 'abnormal' data or actions, that require further attention or investigation by the data reviewer.

Reviewers should have sufficient knowledge and system access to review relevant audit trails, raw data and metadata (see also 'data governance').

Where systems do not meet the audit trail and individual user account expectations, demonstrated progress should be available to address these shortcomings. This should either be through add-on software that provides these additional functions or by an upgrade to a compliant system. Where remediation has not been identified or subsequently implemented in a timely manner a deficiency may be cited.

6.14. Electronic signatures

A signature in digital form (bio-metric or non-biometric) that represents the signatory. This should be equivalent in legal terms to the handwritten signature of the signatory.

The use of electronic signatures should be appropriately controlled with consideration given to:

- How the signature is attributable to an individual.

¹ It is expected that GMP facilities with industrial automation and control equipment/ systems such as programmable logic controllers should be able to demonstrate working towards system upgrades with individual login and audit trails (reference: Art 23 of Directive 2001/83/EC).



- How the act of 'signing' is recorded within the system so that it cannot be altered or manipulated without invalidating the signature or status of the entry.
- How the record of the signature will be associated with the entry made and how this can be verified.
- The security of the electronic signature i.e. so that it can only be applied by the 'owner' of that signature.

It is expected that appropriate validation of the signature process associated with a system is undertaken to demonstrate suitability and that control over signed records is maintained. Where a paper or pdf copy of an electronically signed document is produced, the metadata associated with an electronic signature should be maintained with the associated document.

The use of electronic signatures should be compliant with the requirements of international standards. The use of advanced electronic signatures should be considered where this method of authentication is required by the risk assessment. Electronic signature or E-signature systems must provide for "signature manifestations" i.e. a display within the viewable record that defines who signed it, their title, and the date (and time, if significant) and the meaning of the signature (e.g. verified or approved).

An inserted image of a signature or a footnote indicating that the document has been electronically signed (where this has been entered by a means other than the validated electronic signature process) is not adequate. Where a document is electronically signed then the metadata associated with the signature should be retained.

For printed copies of electronically signed documents refer to True Copy section.

Expectations for electronic signatures associated with informed consent (GCP) are covered in alternative guidance (MHRA/HRA DRAFT Guidance on the use of electronic consent).

6.15. Data review and approval

The approach to reviewing specific record content, such as critical data and metadata, cross-outs (paper records) and audit trails (electronic records) should meet all applicable regulatory requirements and be risk-based.

There should be a procedure that describes the process for review and approval of data. Data review should also include a risk-based review of relevant metadata, including relevant audit trails records. Data review should be documented and the record should include a positive statement regarding whether issues were found or not, the date that review was performed and the signature of the reviewer.

A procedure should describe the actions to be taken if data review identifies an error or omission. This procedure should enable data corrections or clarifications to provide visibility of the original record, and traceability of the correction, using ALCOA principles (see 'data' definition).

Where data review is not conducted by the organisation that generated the data, the responsibilities for data review must be documented and agreed by both parties. Summary

7 PRINCIPLE

- 8 4.1. Documentation constitutes an essential part of the quality assurance system and is key to op-
9 erating in compliance with GMP requirements. The various types of documents and means
10 used should be fully understood and defined in the regulated user's pharmaceutical quality
11 system.
- 12 4.2. It should be determined by the regulated user which legal provisions apply to documentation
13 considering new technologies, hybrid solutions and services used.
- 14 4.3. Appropriate documentation practices should be applied with respect to the type of document
15 regardless of the applied technology or service used.
- 16 4.4. The present document was supplemented by requirements regarding new technologies, hybrid
17 solutions and new services, whereby a risk-based approach as element of a data governance
18 system is considered pivotal for scalability of integrity control measures.
- 19 4.5. Quality risk management principles should be applied to ensure that the pharmaceuti-
20 cal quality system includes sufficient instructional details. It should facilitate a common
21 understanding of the requirements. In addition to providing for recording of the various pro-
22 cesses and risk-based evaluation of any observations, it should demonstrate the ongoing appli-
23 cation of all requirements.
- 24 4.6. Suitable controls should be implemented using a risk-based approach to ensure the accuracy,
25 integrity, availability, and legibility of documents. Documents should be free from errors and
26 available in human readable form.
- 27 4.7. Documentation may exist in a variety of forms, including paper-based, electronic, or other
28 means (e.g. photography, imagery, video and audio recordings). The main objective of the
29 documentation system is to establish, control, monitor and record all activities which directly
30 or indirectly impact on all aspects of the quality and safety of medicinal products, using a risk-
31 based approach.
- 32 4.8. Whether documents are created, stored, and managed electronically, paper based, by other
33 means or through a hybrid system, they must meet the same GMP requirements for legibility,
34 accuracy, integrity, and completeness throughout the whole lifecycle. This also applies when
35 documentation is outsourced.
- 36 4.9. There are two primary types of documentation used to manage and demonstrate GMP compli-
37 ance:
 - 38 i. instructions (directions, requirements) and
 - 39 ii. records/reports.

40 DATA GOVERNANCE SYSTEMS

- 41 4.10. Regardless of how documents are created, handled, stored, and managed (i.e., using elec-
42 tronic, paper-based, or hybrid systems), the regulated user should establish a data governance
43 system integral to the pharmaceutical quality system to define, prioritise and communicate
44 their data integrity risk management activities. Arrangements for data governance should be
45 documented and reviewed regularly.
- 46 4.11. Regulated users should design and operate a data governance system which provides an ac-
47 ceptable state of control based on the risk assessment, which is documented with supporting
48 rationale. A data governance system should be consistent with the principles of quality risk
49 management.
- 50 4.12. To ensure integrity of data the governance system should cover the entire data lifecycle and
51 ensure controls commensurate with the principle of quality risk management. The data lifecy-
52 cle should refer to:
 - 53 i. Creation and recording of data.
 - 54 ii. Processing of (raw) data to reported (derived) data.

- iii. Verification of completeness, consistency and accuracy of all data (raw and derived data). For derived data the traceability which allows reconstruction of all data processing activities should be maintained.
 - iv. Decision making relying on data (or derived data).
 - v. Retaining, archiving and retrieval of data. To protect it from loss or unauthorised alteration it should be commensurate with the principles of quality risk management.
 - vi. Retirement or destruction of data at the end of the lifecycle in a controlled manner.
- 4.13. Data governance systems should rely on a risk management approach and consider:
- i. Data criticality (impact to decision making and product quality) and
 - ii. Data risk (opportunity for data alteration and deletion, and likelihood of detection / visibility of changes by the regulated user's routine review processes).
- 4.14. Data governance systems should rely on the incorporation of suitably designed systems, the use of technologies and data security measures, combined with specific expertise to ensure that data integrity is effectively controlled over the data lifecycle.
- 4.15. Data governance systems should address data ownership throughout the entire lifecycle.
- 4.16. Data governance systems should consider the design, operation and monitoring of processes and systems to comply with the principles of data integrity.
- 4.17. Data governance systems should consider risk mitigation. The effectiveness of risk mitigation measures should be reviewed regularly, regardless of whether they are temporary or permanent. Residual risks should be reviewed periodically and communicated to management.
- 4.18. Data governance systems should consider and ensure the periodic review of service provider's data management policies and risk control strategies intended to minimise potential risks to data integrity. The frequency of such reviews should be based on the criticality of the services provided, using risk management principles.

RISK MANAGEMENT

- 4.19. The regulated user should adopt a risk-based approach in documentation throughout the entire lifecycle of data, regardless of the technology, hybrid solution or service used and should demonstrate an understanding for data criticality, data risk and data quality.
- 4.20. Controls over the data lifecycle should be established which are commensurate with the principles of quality risk management. The depth of data governance and risk management activities should be justified and commensurate with the risks to product quality and patient safety.
- 4.21. Decisions on the extent of measures to ensure data integrity should be based on a documented rationale and documented risk assessment taking into consideration data criticality and data risk.
- 4.22. Irrespective of processes used to generate electronic data, they must be included in the requirements for the qualification or validation of the relevant computerised systems according to Annex 11.

GENERAL REQUIREMENTS FOR DOCUMENTATION

- 4.23. The pharmaceutical quality system should describe all documents required to ensure product quality and patient safety. Documents may be created, recorded, provided, approved, communicated, stored, and archived electronically, paper based or in a hybrid system. The reliance on electronic, paper-based or different means, hybrid solutions or hosted services in maintenance and retention of documentation requires the compliance with all EU GMP provisions including Annex 11 if decision making in manufacturing (e.g. batch release based on in-process controls and process analytical technologies) is supported by automatic validation scripts or artificial intelligence (Annex 22).
- 4.24. The accountability for the integrity of documents, records or (raw) data produced or processed

with artificial intelligence or any other automatic means (e.g. validation scripts) rests with the regulated user.

4.25. The support by any automatic means (e.g. validation scripts or artificial intelligence) should be included in a pharmaceutical quality system regardless of the service located on premise or as a hosted service. The records created electronically should enable a trend analysis of quality-critical data.

4.26. To ensure data integrity, data which is recorded or processed electronically should not be converted to or stored in a paper form unless it meets the requirements set out in section 13 “hybrid systems” or the conversion is validated or verified for accuracy.

MASTER DOCUMENTS

4.27. Specifically required master documents (not exhaustive list):

i. **Site Master File:** Refer to EU GMP Guidelines, Volume 4 “Explanatory Notes on the preparation of a Site Master File”.

ii. **Validation Master Plan:** A document describing the key elements of the site qualification and validation program. Master documents should be evaluated and reviewed on a regular basis.

iii. **Instructions** (directions, or requirements) type:

- **Specification:** Refer to glossary for definition

- **Manufacturing Formulae, Processing, Packaging and Testing Instruction:** Provide complete detail on all the starting materials, equipment, and computerised systems (if any) to be used and specify all processing, packaging, sampling, and testing instructions to ensure batch to batch consistency. In-process controls and process analytical technologies to be employed should be specified where relevant, together with acceptance criteria.

- **Procedures:** (Otherwise known as Standard Operating Procedures, or SOPs), documented set of instructions for performing and recording operations.

- **Protocol:** defined set of activities to provide instructions for performing and recording certain discreet operations.

- **Technical / Quality Agreement:** Written proof of agreement between contract givers and acceptors for outsourced activities.

iv. **Record/Report type:**

- **Record:** Provide evidence of various actions taken to demonstrate compliance with instructions, e.g. activities, events, investigations, and in the case of manufactured batches a history of each batch of product, including its distribution. Records include the raw data which is used to generate other records. For electronic records regulated users should define which data are to be used as raw data. At least, all data on which quality decisions are based should be defined as raw data.

Records can also exist as hybrid records which is a combination of paper records, electronic records or by other means. The completeness and integrity of records, including all relevant raw data and meta data should be ensured and protected based on risk.

- **Certificate of Analysis:** Provide a summary of testing results on samples of products or materials¹ together with the evaluation for compliance to a stated specification.

¹ Alternatively, the certification may be based, in-whole or in-part, on the assessment of real time data (summaries and exception reports) from batch related process analytical technology (PAT), parameters or metrics as per the approved marketing authorisation dossier.

185 mention should be made of any substance that may disappear while processing, of
186 processing aids needed or any other material relevant for product quality.

187 iv. A statement of the expected final yield with the acceptable limits, and of relevant in-
188 termediate yields, where applicable.

189 4.34. The Processing Instructions should include:

190 i. A statement of the processing location and the principal equipment to be used.

191 ii. The methods, or reference to the methods, to be used for preparing the critical equip-
192 ment (e.g. cleaning, assembling, calibrating, sterilising).

193 iii. Checks that the equipment and workstation are clear of previous products, docu-
194 ments or materials not required for the planned process, and that equipment is clean
195 and suitable for use.

196 iv. Detailed stepwise processing instructions [e.g. checks on materials, pre-treatments,
197 sequence for adding materials, critical process parameters (time, temp etc)].

198 v. The instructions for any in-process controls with their limits.

199 vi. Where necessary, the requirements for bulk storage of the products; including the
200 container, labelling and special storage conditions where applicable.

201 vii. Any special precautions to be observed.

202 **Packaging Instructions**

203 4.35. Approved Packaging Instructions for each product, pack size and type should exist. These
204 should include, or have a reference to, the following:

205 i. Name of the product; including the batch number of bulk and finished product.

206 ii. Description of its pharmaceutical form, and strength where applicable.

207 iii. The pack size expressed in terms of the number, weight or volume of the product in
208 the final container.

209 iv. A complete list of all the packaging materials required, including quantities, sizes
210 and types, with the code or reference number relating to the specifications of each
211 packaging material.

212 v. Where appropriate, an example or reproduction of the relevant printed packaging
213 materials, and specimens indicating where to apply batch number references, and
214 shelf life of the product.

215 vi. Checks that the equipment and workstation are clear of previous products, docu-
216 ments or materials not required for the planned packaging operations (line clear-
217 ance), and that equipment is clean and suitable for use.

218 vii. Checks on functioning of any electronic code readers, label counters or similar de-
219 vices.

220 viii. Special precautions to be observed, including a careful examination of the area and
221 equipment in order to ascertain the line clearance before operations begin.

222 ix. A description of the packaging operation, including any significant subsidiary opera-
223 tions, and equipment to be used.

224 x. Details of in-process controls with instructions for sampling and acceptance limits.

225 **Batch Processing Record**

226 4.36. A Batch Processing Record should be kept for each batch processed. It should be based on

the relevant parts of the currently approved Manufacturing Formula and Processing Instructions, and should contain the following information:

- i. The name and batch number of the product.
- ii. Dates and times of commencement, of significant intermediate stages and of completion of production.
- iii. Identification (initials) of the operator(s) who performed each significant step of the process and, where appropriate, identification (initials) of the person who checked these operations.
- iv. The batch number and/or analytical control number as well as the quantities of each starting material weighed (including the batch number and amount of any recovered or reprocessed material added).
- v. Any relevant processing operation or event and major equipment used.
- vi. A record of the in-process controls and the initials of the person(s) carrying them out, and the results obtained.
- vii. The product yield obtained at different and pertinent stages of manufacture.
- viii. Notes on special problems including details, with signed authorisation for any deviation from the Manufacturing Formula and Processing Instructions.
- ix. Approval by the person responsible for the processing operations.

Note: Where a validated process is continuously monitored and controlled, then automatically generated reports may be limited to compliance summaries and exception/ out-of-specification (OOS) data reports.

With regards to decision making in manufacturing supported by automatic validation scripts or artificial intelligence refer to paragraph 4 of this document.

Batch Packaging Record

4.37. A Batch Packaging Record should be kept for each batch or part batch processed. It should be based on the relevant parts of the Packaging Instructions.

4.38. The batch packaging record should contain the following information:

- i. The name and batch number of the product.
- ii. The date(s) and times of the packaging operations.
- iii. Identification (initials) of the operator(s) who performed each significant step of the process and, where appropriate, the name of any person who checked these operations.
- iv. Records of checks for identity and conformity with the packaging instructions, including the results of in-process controls.
- v. Records of checks that the equipment and workstation are clear of previous products, documents or materials not required for the planned packaging operations (line clearance), that equipment is clean and suitable for use, and that any electronic code readers, label counters or similar devices are functioning as expected.
- vi. Details of the packaging operations carried out, including references to equipment and the packaging lines used.
- vii. Whenever possible, samples of printed packaging materials used, including specimens of the batch coding, expiry dating and any additional overprinting.
- viii. Notes on any special problems or unusual events including details, with signed

- 270 authorisation for any deviation from the Packaging Instructions.
- 271 ix. The quantities and reference number or identification of all printed packaging mate-
272 rials and bulk product issued, used, destroyed or returned to stock and the quantities
273 of obtained product, to provide for an adequate reconciliation. Where there are vali-
274 dated electronic controls in place during packaging there may be justification for not
275 including this information
- 276 x. Approval by the person responsible for the packaging operations.

277 **Receipt**

- 278 4.39. There should be written procedures and records for the receipt of each delivery of each starting
279 material, (including bulk, intermediate or finished goods), primary, secondary and printed
280 packaging materials and QC-samples. The records of the receipts should include:
- 281 i. The name of the material on the delivery notes and the containers.
- 282 ii. The "in-house" name and/or code of material (if different from a).
- 283 iii. Date of receipt.
- 284 iv. Supplier's name and, manufacturer's name.
- 285 v. Manufacturer's batch or reference number.
- 286 vi. Total quantity and number of containers received.
- 287 vii. The batch number assigned after receipt.
- 288 viii. Any relevant comment.
- 289 ix. If applicable, proof of verification that temperature during transportation were within
290 the approved limit.
- 291 4.40. There should be written procedures for the internal labelling, quarantine and storage of start-
292 ing materials, packaging materials, QC samples and other materials, as appropriate.

293 **Sampling**

- 294 4.41. There should be written procedures for sampling, which include the methods and equipment
295 to be used, the amounts to be taken and any precautions to be observed to avoid contamination
296 of the material or any deterioration in its quality (reference to EU GMP Guideline Volume 4,
297 Chapter 6 "Quality Control").

298 **Testing**

- 299 4.42. There should be written procedures for testing materials and products at different stages of
300 manufacture, describing the methods and equipment to be used. The tests performed should
301 be recorded (reference to EU GMP Guideline Volume 4, Chapter 6 "Quality Control").

302 **Other**

- 303 4.43. Written release and rejection procedures should be available for materials and products, and
304 in particular for the certification for sale of the finished product by the Qualified Person(s).
305 All records should be available to the Qualified Person at the time of the release decision. A
306 system should be in place to indicate special observations and any changes to critical data.
- 307 4.44. Records should be maintained for the distribution of each batch of a product in order to facil-
308 itate recall of any batch, if necessary.
- 309 4.45. There should be written policies, procedures, protocols, reports and the associated records of
310 actions taken, or conclusions reached, where appropriate, for GMP relevant actions, including

but not limited to the following examples:

- i. Validation and qualification of processes, equipment and systems.
 - ii. Equipment assembly and calibration.
 - iii. Data integrity.
 - iv. Technology transfer.
 - v. Maintenance, cleaning and sanitation.
 - vi. Personnel matters including signature lists, training in GMP and technical matters, clothing and hygiene and verification of the effectiveness of training.
 - vii. Environmental monitoring.
 - viii. Pest control.
 - ix. Complaints.
 - x. Recalls.
 - xi. Returns.
 - xii. Change control.
 - xiii. Investigations into deviations.
 - xiv. non-conformances e.g. out of specifications.
 - xv. Internal quality/GMP compliance audits.
 - xvi. Summaries of records where appropriate (e.g. product quality review).
 - xvii. Supplier audits.
- 4.46. Clear operating procedures should be available for major items of manufacturing and test equipment.
- 4.47. Logbooks should be kept for major or critical analytical testing, production equipment, and areas where product has been processed or handled. They should be used to record in chronological order, as appropriate, any use of the area, equipment/method, calibrations, maintenance, cleaning or repair operations, including the dates and identity of people who carried these operations out.
- 4.48. An inventory of documents within the pharmaceutical quality system should be maintained.

GENERATION AND CONTROL OF DOCUMENTATION

- 4.49. All types of documents (instructions and/or records) should be defined and adhered to regardless of the documentation technology, hybrid solution or service. The technology, hybrid solution or provided service needs to be understood regardless of complexity, should be adequately documented, and validated with risk-based controls in place. Relationships and control measures for master documents, official copies, data handling and records need to be defined for both hybrid and homogenous systems regardless of the type of service. Appropriate controls for documents should be implemented to ensure the completeness, integrity and legibility of the records throughout the lifecycle.
- 4.50. Documents should be designed, prepared, reviewed, and distributed in a controlled manner. They should comply with the relevant parts of Product Specification Files, Manufacturing and Marketing Authorisation dossiers, or dossiers of Investigational Medicinal Products, as appropriate. The reproduction of working documents from master documents should not allow any error or alteration to be introduced through

the reproduction process.

4.51. Documents should be regularly reviewed and kept up to date. Documents should be approved, signed, and dated by appropriate and authorised personnel. Documents should have unambiguous contents and be uniquely identifiable. The effective date should be defined.

4.52. Documents containing instructions should be laid out in an orderly fashion and be easy to review. The style and language of documents should fit with their intended use. Standard operating procedures, work instructions and methods should be written in an imperative mandatory style by using predefined format. Data entry formats for completion of documents should be clearly defined. Written instructions may be supported with pictures, photos or videos. The documents containing the instructions should be easily accessible at the place where the described activities are carried out.

4.53. Instructions and procedures within the Quality Management System should be regularly reviewed and kept updated.

4.54. The issuance, revision, superseding and withdrawal of all documents should be controlled with maintenance of revision histories.

4.55. Hand-written instructions are discouraged. Where documents require the manual entry of data, sufficient space should be provided for such entries to ensure adequately clear and legible manual recording.

GOOD DOCUMENTATION PRACTICE

4.56. Good Documentation practices are key to ensuring data integrity, and a fundamental part of a well-designed pharmaceutical quality system.

4.57. Procedures outlining good documentation practices and arrangements for document control should be available within the pharmaceutical quality system. Good documentation practices should be implemented and enforced to ensure data integrity.

4.58. Data entries should be accurate, and made in clear, legible, indelible way. Recorded media should be durable throughout the retention period. If this is not feasible, then true copies should be generated. For this case a documented system should be in place to verify and record the integrity of the copy.

4.59. Records should be made or completed at the time each action is taken and in such a way that all GMP activities are traceable. It should be possible to identify the individual or the system that performed the task and when the task was performed.

4.60. Any alteration made to the entry on a document should be signed by the individual who made the change and dated; the alteration should permit the reading of the original information. Where appropriate, the reason for the alteration should be recorded.

4.61. Records need to be a truthful and consistent representation of facts to ensure the accuracy of information, including data that is used to make critical decisions about the quality of products.

4.62. Specific controls should be implemented to ensure the integrity of raw data and results recorded on paper. These may include, but are not limited to:

i. control over the issuance and use of loose paper sheets (blank forms) at the time of recording data.

ii. control over the issuance of bound, paginated notebooks.

iii. control over the issuance and reconciliation of sequentially numbered copies of

blank forms with authenticity controls.

- iv. Control that raw data is contemporaneously and accurately recorded by permanent means.

4.63. Basic data integrity principles (table 1) applicable to both paper and electronic systems (i.e. ALCOA ++):

Table 1: Data integrity principles

Attributes	Requirement
Attributable	It should be possible to identify the individual or computerised system that performed a recorded task and when the task was performed. This also applies to any changes made to records, such as corrections, deletions, and changes where it is important to know who made a change, when, and why.
Legible	All records should be legible – the information should be readable and unambiguous in order to be understandable and of use. This applies to all information that would be required to be considered complete, including all original records or entries. Where the ‘dynamic’ nature of electronic data (the ability to search, query, trend, etc.) is important to the content and meaning of the record, the ability to interact with the data using a suitable application is important to the ‘availability’ of the record.
Contemporaneous	The evidence of actions, events or decisions should be recorded as they take place. This documentation should serve as an accurate attestation of what was done, or what was decided and why, i.e. what influenced the decision at that time.
Original	The original record can be described as the first capture of information, whether recorded on paper (static) or electronically (usually dynamic, depending on the complexity of the system). Information that is originally captured in a dynamic state should remain available in that state.
Accurate	Records need to be a truthful representation of facts to be accurate. Ensuring records are accurate is achieved through many elements of a robust pharmaceutical quality system. This can be comprised of: • equipment related factors such as qualification, calibration, maintenance, and computer validation. • policies and procedures to control actions and behaviours, including data review procedures to verify adherence to procedural requirements.

	• deviation management including root cause analysis, impact assessments and CAPA. • trained and qualified personnel who understand the importance of following established procedures and documenting their actions and decisions. Together, these elements aim to ensure the accuracy of information, including scientific data that is used to make critical decisions about the quality of products.
Complete	All information that would be critical to recreating an event is important when trying to understand the event. It is important that information is not lost or deleted. The level of detail required for an information set to be considered complete would depend on the criticality of the information. A complete record of data generated electronically includes relevant metadata.
Consistent	Information should be created, processed, and stored in a logical manner that has a defined consistency. This includes policies or procedures that help control or standardize data (e.g. chronological sequencing, date formats, units of measurement, approaches to rounding, significant digits, etc.).
Enduring	Records should be kept in a manner such that they exist for the entire period during which they might be needed. This means they need to remain intact and accessible as an indelible/durable record throughout the record retention period.
Available	Records should be available for review at any time during the required retention period, accessible in a readable format to all applicable personnel who are responsible for their review whether for routine release decisions, investigations, trending, annual reports, audits or inspections.
Traceable	Traceability is the ability to trace the history, modification or location of data by means of recorded identifications.

404

405 SIGNATURES IN GMP RELEVANT DOCUMENTATION

406 4.64. Signatures are essential for ensuring accountability for activities in a GMP environ-
407 ment at the time points the signatures are executed.

408 4.65. A signature represents the legally binding will of the signatory. The signatory should
409 sign with date and time. If signatures by initials are in use a procedure defining abbrevi-
410 ated signatures should be in place.

- 4.66. The identification of the signatory should be possible. Data or documents which are associated with the signature should be clearly identified. The meaning of the signature (such as review, approval, responsibility, or authorship) associated with the signature should be clear.
- 4.67. The regulated user should establish a signature policy to ensure the adequate application of signatures. Personnel authorised to sign should be clearly identified by name and bound by name to the signature policy.
- 4.68. The regulated user should have identified those records which require a legally binding signature.
- 4.69. Signatures should be indisputable and traceable to the signatory and the signed document or record, regardless, if a signature is applied on paper or electronically.
- 4.70. If records exist electronically such records should be signed electronically. The use of a hybrid system should be avoided. If signatures exist parallel in paper and electronically (e.g. in hybrid systems), the regulatory relevant signature should be defined by the regulated user.
- 4.71. The signatory should be qualified and authorised to perform the relevant tasks or reviews.
- 4.72. The regulated user should define the signatory's role and responsibility in the signation process.
- 4.73. The regulated user should ensure that the signatory's role and qualification is consistent with the meaning (intent) of a signature.
- 4.74. To ensure the integrity of signatures during the whole life cycle of data the regulated user should establish the management and control of signatures as an element of a data governance system.
- 4.75. The data or documents which the signature is relevant for should fulfil the ALCOA++ principles.

RETENTION OF DOCUMENTS

- 4.76. It should be clearly defined which record is related to each activity and where this record is located, regardless of the technology, hybrid solution or service used. Risk-based control methods should be in place to ensure the integrity of the record throughout the lifecycle. The control measures should be covered by the validation scope. In case of electronic recording such measures should include back-up, restore and archiving procedures as well as physical and logical controls. If the regulated user relies on hosted services, it is the responsibility of the regulated user to understand, approve and justify the control measures of the hosted service provider based on a service level agreement. Records should be available for review at any time during the required retention period, accessible in a human readable format to all applicable personnel.
- 4.77. Specific requirements apply to batch documentation which must be kept for one year after expiry of the batch to which it relates or at least five years after certification of the batch by the Qualified Person, whichever is the longer. For investigational medicinal products, the batch documentation must be kept for at least five years after the completion or formal discontinuation of the last clinical trial in which the batch was used. Other requirements for retention of documentation may be described in legislation in relation to specific types of products (e.g. Advanced Therapy Medicinal

Products and products derived from human blood or human plasma) and specify that longer retention periods be applied to certain documents.

- 4.78. For other types of documentation, the retention period will depend on the business activity which the documentation supports. Critical documentation, including raw data (for example relating to validation or stability), which supports information in the Marketing Authorisation should be retained whilst the authorization remains in force. It may be considered acceptable to retire certain documentation (e.g. raw data supporting validation reports or stability reports) where the data has been superseded by a full set of new data. Justification for this should be documented and should consider the requirements for retention of batch documentation; for example, in the case of process validation data, the accompanying raw data should be retained for a period at least as long as the records for all batches whose release has been supported on the basis of that validation exercise.
- 4.79. A documented process for the disposal of records should be in place to ensure that the correct original records or true copies are disposed only after the defined retention period. Measures should be in place to reduce the risk of deleting the wrong documents. The access rights allowing disposal of records should be controlled.

DATA INTEGRITY IN DOCUMENTATION

- 4.80. The method of documentation should be integrated in the regulated user's pharmaceutical quality system. Documents or records should be controlled in a risk-based approach regardless of whether located in-house or in the form of hosted services. The regulated user should apply the principles of data integrity, data criticality and data risk within a data governance system and should consider the complete lifecycle of data. The data governance system should be an element of the pharmaceutical quality system. The ownership of data and the responsibility for data integrity should be defined.
- 4.81. Risk based control measures should be commensurate with the type and the complexity of a system. The pharmaceutical quality system should interface with independent review practices to detect risks to data integrity. Risks from human factors should be considered for effectively ensuring data integrity Risk-reducing measures such as second person oversight, verification and checks should be implemented where appropriate and in the appropriate time to ensure critical process and testing steps are accurately and contemporaneously recorded.

HYBRID SYSTEMS

- 4.82. They should be clearly defined and identified, and each contributing element of the system validated and controlled according to risk management principles.
- 4.83. A detailed description of the entire system should be available. The description should outline all major components, their functions, and interactions with each other as well as control for data management and data integrity. Procedures and records should be available to manage and appropriately control the interface between manual and computerised systems.
- 4.84. Appropriate quality risk management principles should be followed when assessing, defining, and demonstrating the effectiveness of control measures applied to the system.
- 4.85. Procedures should be in place to manage the review of data generated by hybrid

501 systems which clearly outline the process for the evaluation, approval and archiving
502 of electronic and paper-based data.

503 GLOSSARY

504 ALCOA++ An acronym for “attributable, legible, contemporaneous, original and accu-
505 rate”, which puts additional emphasis on the attributes complete, consistent,
506 enduring, available and traceable – implicit basic ALCOA principles.

507 Archiving Long term, or permanent retention of completed documentation and relevant
508 metadata in its final form for the purposes of reconstruction of a process or
509 activity.

510 Automated script A piece of code used to automate repetitive processes

511 Data The contents of the record. Data may be defined as measurable or descriptive
512 attribute of a physical entity, process, or event.

513 Data governance The total sum of arrangements to ensure that data, irrespective of the format in
514 which it is generated, recorded, processed, retained and used, will be attribut-
515 able, legible, contemporaneous, original, accurate, complete, consistent, en-
516 during, and available throughout the data lifecycle.

517 Data integrity Data integrity refers to the completeness, consistency, and accuracy of data.
518 Complete, consistent, and accurate data should be attributable, legible, con-
519 temporaneously recorded, original or a true copy, accurate and traceable (AL-
520 COA++).

521 Data lifecycle All processes related to the creating, recording, processing, reviewing, chang-
522 ing, analysing, reporting, transferring, storing, migrating, archiving, retrieving,
523 and deleting of data.

524 Data management The set of all methodological, conceptual, organisational and technical
525 measures and procedures for handling data with the aim of incorporating it into
526 business processes.

527 Data risk The combination of the probability of occurrence of harm and the severity of
528 that harm related to data (incompleteness, alterations or loss which compro-
529 mise the integrity of data).

530 Data criticality The degree of influence that data have on product safety as well as the regula-
531 tory compliance of processes, decisions and product quality.

532 Data Risk Management An activity to be applied throughout the lifecycle of data considering the need
533 to ensure data integrity. Risk management consists of risk identification, risk
534 assessment, risk mitigation and risk control. Risk management should link to
535 other relevant procedures (e.g. configuration and change management, man-
536 agement processes for data, business risks, etc.).

537 Data Risk Assessment The process of evaluating the risks associated with the regulated user’s data. It
538 ensures an efficient and effective approach to data integrity by considering the
539 vulnerability of data to involuntary or deliberate alteration resulting in risk-
540 based control measures.

541	Data ownership	The allocation of responsibilities for control of data to a specific process
542		owner. Companies should implement systems to ensure that responsibilities for
543		systems and their data are appropriately allocated, and responsibilities under-
544		taken.
545	Data quality	The degree to which a set of inherent characteristics (quality dimensions) of
546		data fulfils requirements.
547		Data should be fit for use in their intended operational, decision-making, and
548		other roles and should exhibit conformance to regulatory standards that have
549		been set, so that fitness for use is achieved.
550	Document	A formatted compilation of data. Operations and activities that are memorial-
551		ized in (electronic) records may consist of one or more documents that describe
552		the activity in a moment of time.
553	Electronic record	Any combination of text, graphics, data, audio, pictorial, or other information
554		representation in digital form that is created, modified, maintained, archived,
555		retrieved, or distributed by a computer system.
556	Hybrid system	A combination of paper based and electronic means.
557	Homogenous systems	A system that is either paper or electronically based on-premises or a cloud
558		service.
559	Meta data	Describe the attributes of data and provides context and meaning. Metadata is
560		any information used for the identification, description, and relationships of
561		electronic records or their elements. Metadata gives data meaning, provides
562		context, defines structure, and enables retrievability across systems, and usa-
563		bility, authenticity, and auditability across time.
564	Raw Data	Raw data is defined as the original record (data) which can be described as the
565		first capture of stored information, whether recorded on paper or electronically.
566		Information that is originally captured in a dynamic state should remain avail-
567		able in that state.
568	Record	Memorializes, or makes information permanent about, an action, activity and
569		event that caused its creation.
570	Regulated user	Marketing Authorisation Holder, Manufacturers, control laboratories, import-
571		ers, and wholesale distributors (if the wholesale distributor holds a manufac-
572		turing license).
573	Risk based approach for data integrity	
574		A process to define critical data, documents and the actions used to monitor
575		activities like capturing, derivation, migration, storage, communication and ar-
576		chiving to ensure that data and documents remain in a state of control through-
577		out the entire lifecycle and to maintain its integrity.
578	Specification	A list of tests, references to analytical procedures, and appropriate acceptance
579		criteria that are numerical limits, ranges, or other criteria for the test described.
580		It establishes the set of criteria to which a material should conform to be con-
581		sidered acceptable for its intended use. "Conformance to specification" means
582		that the material, when tested according to the listed analytical procedures, will
583		meet the listed acceptance criteria.

584	True copy	An exact copy of original documentation that preserves the same content,
585		meaning and attributes of the original. The term “true copy” is synonymous
586		with “certified” or “verified copy”.
587	Type of service	On-premises IT service or outsourced hosted (cloud) IT service
588	Verified copy	Refer to definition of true copy
589	Data Risk Assessment	The process of evaluating the risks associated with the regulated user’s data.